MACROEVOLUTION

# Module map

- Module 14: Macroevolution
- Connected lab: lab-15_macroevolution.md
- Use this deck as the visual path through the module's evidence and retrieval work.

## Module 14
Macroevolution

1. Species concepts
2. Reproductive isolation
3. Prezygotic barriers
4. Postzygotic barriers
5. Speciation patterns and phylogeny

**Lab evidence**
lab-15_macroevolution.md

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# Learning objectives

- State the biological species concept and its limits.
- Explain reproductive isolation as reduced gene flow.
- Classify examples as prezygotic or postzygotic barriers.
- Compare allopatric and sympatric speciation.
- Interpret speciation as population-level divergence over time.

**OBJECTIVE 1**
State the biological species concept and its limits.

**OBJECTIVE 2**
Explain reproductive isolation as reduced gene flow.

**OBJECTIVE 3**
Classify examples as prezygotic or postzygotic barriers.

**OBJECTIVE 4**
Compare allopatric and sympatric speciation.

**OBJECTIVE 5**
Interpret speciation as population-level divergence over time.

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# Topic sequence

- Species concepts: The biological species concept uses natural interbreeding, viable fertile offspring, and isolation. It is difficult to apply to fossils or asexual organisms.
- Reproductive isolation: Reproductive isolation reduces gene flow. Appearance or separation alone does not establish species boundaries; some distinct species hybridize.
- Prezygotic barriers: Prezygotic barriers act before fertilization: habitat, timing, behavior, anatomy, or gamete incompatibility can prevent a zygote.
- Postzygotic barriers: Postzygotic barriers reduce hybrid development, survival, or reproduction. Classify the stage using the evidence provided.
- Speciation patterns and phylogeny: Allopatric speciation begins with separation; sympatric speciation does not. Polyploid lineages must reproduce. Phylogenies show branching ancestry.

01

**Species concepts**

The biological species concept uses natural interbreeding, viable fertile offspring, and isolation. It is difficult to apply to fossils or asexual organisms.

02

**Reproductive isolation**

Reproductive isolation reduces gene flow. Appearance or separation alone does not establish species boundaries; some distinct species hybridize.

03

**Prezygotic barriers**

Prezygotic barriers act before fertilization: habitat, timing, behavior, anatomy, or gamete incompatibility can prevent a zygote.

04

**Postzygotic barriers**

Postzygotic barriers reduce hybrid development, survival, or reproduction. Classify the stage using the evidence provided.

05

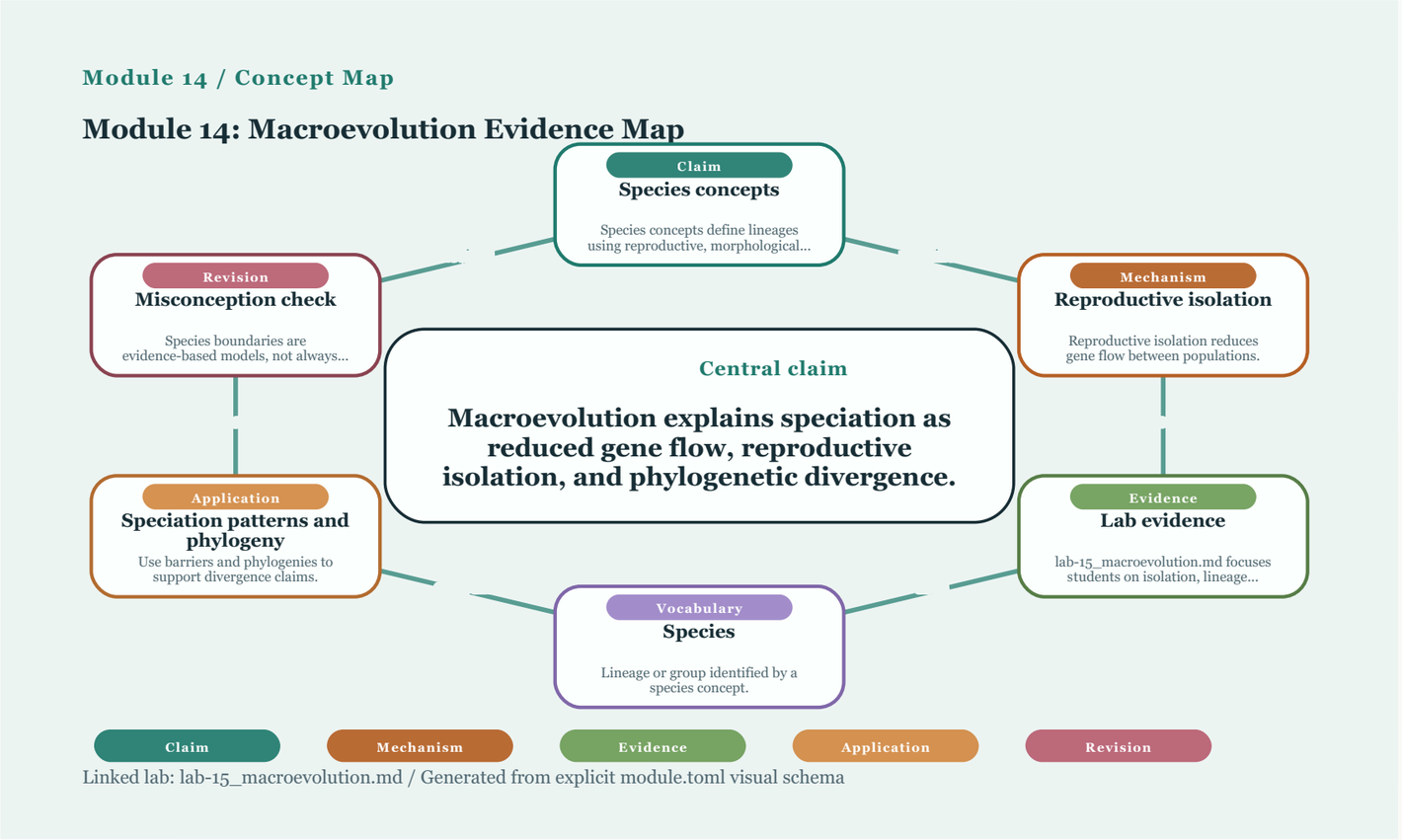
**Speciation patterns and phylogeny**

Allopatric speciation begins with separation; sympatric speciation does not. Polyploid lineages must reproduce. Phylogenies show branching ancestry.

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# Module 14: Macroevolution Evidence Map

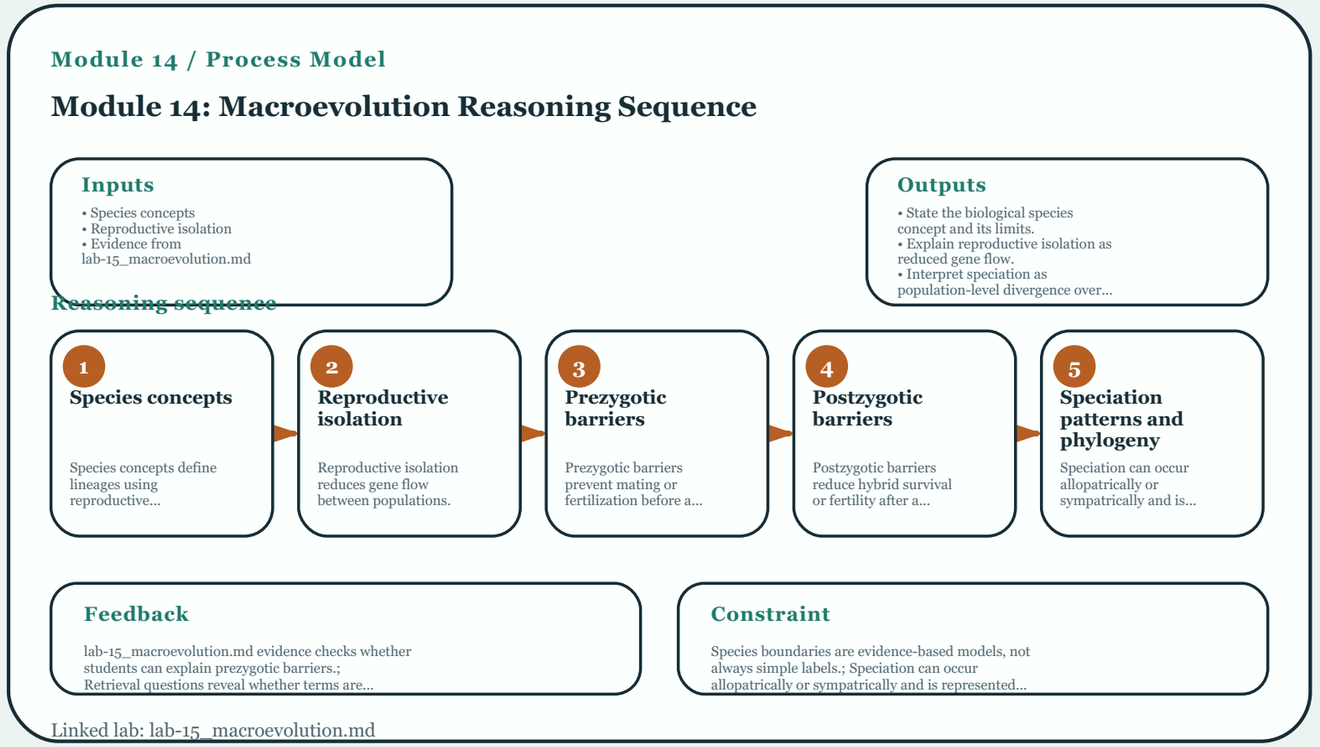
- Module focus: Macroevolution
- Central claim: Macroevolution explains speciation as reduced gene flow, reproductive isolation, and phylogenetic divergence.
- Key nodes to connect: Species concepts, Reproductive isolation, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-15_macroevolution.md supplies an evidence surface for the map.



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# Module 14: Macroevolution Reasoning Sequence

- Module focus: Macroevolution
- Trace the model: Species concepts -> Reproductive isolation -> Prezygotic barriers -> Postzygotic barriers
- Inputs become outputs: Species concepts, Reproductive isolation -> State the biological species concept and its limits., Explain reproductive isolation as reduced gene flow.
- Feedback check: lab-15_macroevolution.md evidence checks whether students can explain prezygotic barriers.
- Lab connection: lab-15_macroevolution.md gives students a process to observe or test.



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# Terms as evidence handles

- Species: Lineage or group identified by a species concept.
- Biological species concept: Species defined by interbreeding and reproductive isolation.
- Reproductive isolation: Barriers that reduce gene flow between groups.
- Prezygotic barrier: Barrier before fertilization.
- Postzygotic barrier: Barrier after fertilization.
- Allopatric speciation: Speciation with geographic separation.

**SPECIES**
Lineage or group identified by a species concept.

**BIOLOGICAL SPECIES CONCEPT**
Species defined by interbreeding and reproductive isolation.

**REPRODUCTIVE ISOLATION**
Barriers that reduce gene flow between groups.

**PREZYGOTIC BARRIER**
Barrier before fertilization.

**POSTZYGOTIC BARRIER**
Barrier after fertilization.

**ALLOPATRIC SPECIATION**
Speciation with geographic separation.

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# Lab connection

- Lab file: lab-15_macroevolution.md
- Lab evidence should test or illustrate: Prezygotic barriers
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

**QUESTION**
Species concepts

**EVIDENCE**
lab-15_macroevolution.md

**CLAIM**
State the biological species concept and its limits.

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# Contrast check

- Do not stop at: The biological species concept uses natural interbreeding, viable fertile offspring, and isolation. It is difficult to apply to fossils or asexual organisms.
- Stronger explanation: Allopatric speciation begins with separation; sympatric speciation does not. Polyploid lineages must reproduce. Phylogenies show branching ancestry.
- The goal is to move from a named idea to a testable biological explanation.

**SURFACE**

The biological species concept uses natural interbreeding, viable fertile offspring, and isolation. It is difficult to apply to fossils or asexual organisms.

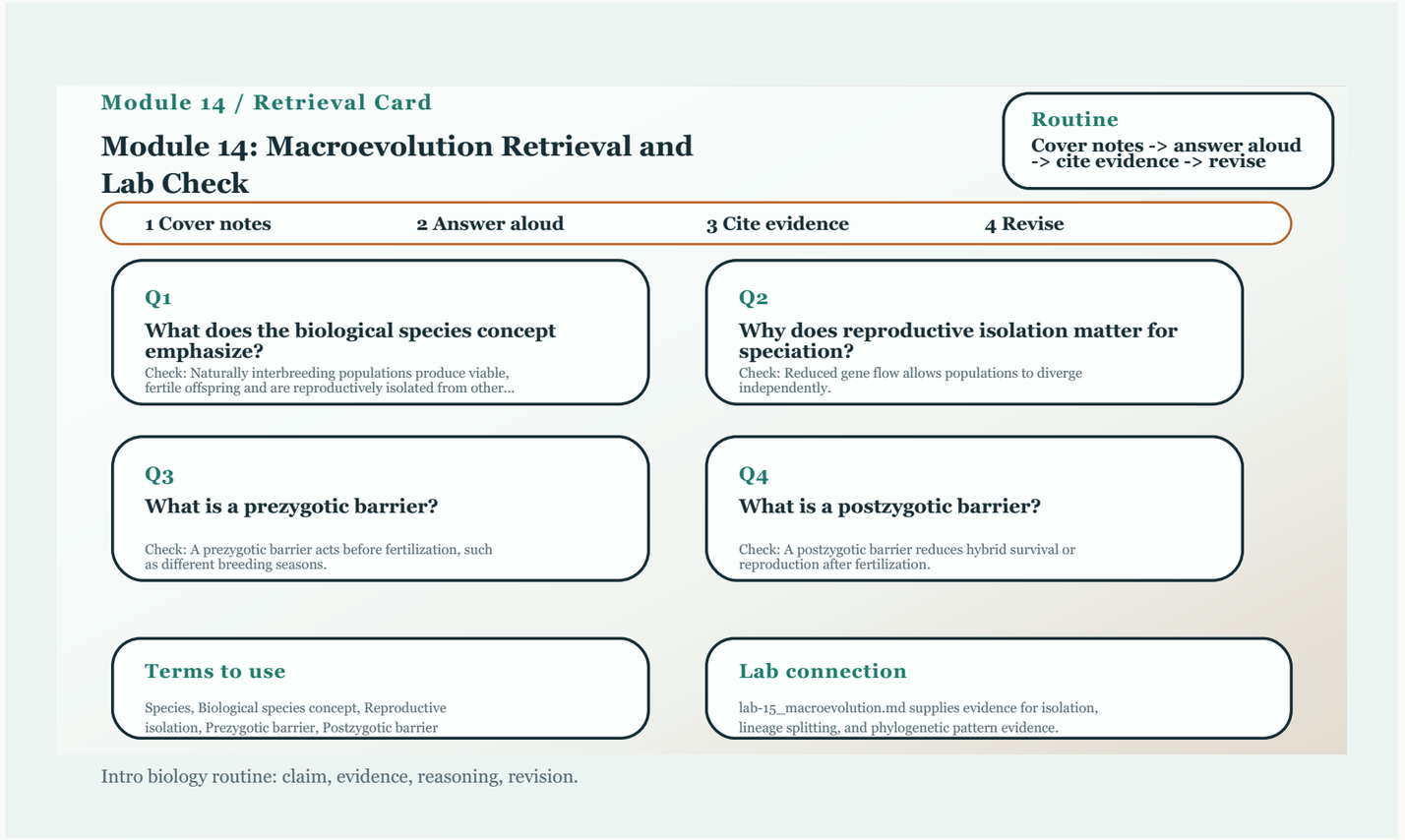
**DEEPER**

Allopatric speciation begins with separation; sympatric speciation does not. Polyploid lineages must reproduce. Phylogenies show branching ancestry.

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# Module 14: Macroevolution Retrieval and Lab Check

- Module focus: Macroevolution
- Retrieval prompt: What does the biological species concept emphasize?
- Answer check: Naturally interbreeding populations produce viable, fertile offspring and are reproductively isolated from other groups.
- Required terms: Species, Biological species concept, Reproductive isolation
- Lab connection: lab-15_macroevolution.md; lab-15_macroevolution.md supplies evidence for isolation, lineage splitting, and phylogenetic pattern evidence.




Intro biology routine: claim, evidence, reasoning, revision.

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# Practice quiz bridge

- 1. Under the biological species concept, two groups count as different species when they:
- 2. Two related species breed at different times of year and so never mate. This is an example of a:
- 3. A mountain range splits one population into two groups that slowly diverge into separate species. This is:
- 4. A mule, the offspring of a horse and a donkey, is healthy but sterile. This illustrates a:

**QUESTION 1**

**Answer first**

Under the biological species concept, two groups count as different species when they:

**QUESTION 2**

**Answer first**

Two related species breed at different times of year and so never mate. This is an example of a:

**QUESTION 3**

**Answer first**

A mountain range splits one population into two groups that slowly diverge into separate species. This is:

**QUESTION 4**

**Answer first**

A mule, the offspring of a horse and a donkey, is healthy but sterile. This illustrates a:

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# Synthesis and exit ticket

- Exit claim: explain species concepts using evidence from lab-15_macroevolution.md.
- Must include: Species, Biological species concept, Reproductive isolation.
- Revision prompt: what evidence would change your explanation?

**CLAIM**
Species concepts

**EVIDENCE**
lab-15_macroevolution.md

**REVISION**
What would change your mind?